

MITRE ATT&CK

Advanced Persistent Threat Hunting & TTP Overlap Analysis

Threat Intelligence Research Report

APT Groups Analysed: APT41 • KIMSUKY • SANDWORM • LAZARUS GROUP

Data Source: SOCRadar.io | Framework: MITRE ATT&CK Enterprise v19

Total APTs Identified: 1,138

Date: 15 June 2026

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1. Executive Summary

This report documents a structured threat hunting engagement conducted against a defined industry vertical and geographic region. Leveraging SOCRadar.io as the primary open-source threat intelligence (OSINT) platform, this research identified 1,138 Advanced Persistent Threat (APT) groups with documented activity relevant to the target scope.

From this dataset, four of the most sophisticated and operationally active nation-state threat actors were selected for deep-dive analysis: APT41, KIMSUKY, SANDWORM, and LAZARUS GROUP. Each group's complete Tactics, Techniques, and Procedures (TTPs) were catalogued using the MITRE ATT&CK Enterprise framework (v19) and subsequently visualised and correlated using the MITRE ATT&CK Navigator.

The overlap analysis reveals significant convergence across all 15 ATT&CK tactic categories, demonstrating that adversaries at this level of sophistication deploy remarkably similar operational playbooks — an insight with significant implications for detection engineering and defensive prioritisation.



2. Project Overview

2.1 Project Objectives

- Identify and catalogue APT groups actively targeting a specific industry and geographic region.
- Conduct structured threat intelligence gathering using SOCRadar.io as the OSINT collection platform.
- Map identified APT TTPs to the MITRE ATT&CK Enterprise framework for standardised analysis.
- Visualise TTP coverage and cross-group overlaps using the MITRE ATT&CK Navigator.
- Produce actionable intelligence to support detection engineering and threat-informed defence.

2.2 Scope Definition

Industry Vertical	Energy / Critical Infrastructure (representative of high-value nation-state targets)
Geographic Region	Asia-Pacific (APAC) with secondary coverage of Eastern Europe and North America
Framework Version	MITRE ATT&CK Enterprise v19
Intelligence Platform	SOCRadar.io — Open Source Threat Intelligence & Attack Surface Management
Total APTs in Scope	1,138 threat actors with documented activity matching the defined industry and region
Deep-Dive Selection	APT41, KIMSUKY, SANDWORM, LAZARUS GROUP (selected by operational frequency and sophistication)

2.3 Research Questions

- Which APT groups pose the most significant risk to this industry and region?
- What TTPs do these groups share, and what does convergence indicate about adversary behaviour?
- Which ATT&CK techniques appear consistently across multiple nation-state actors, indicating highest-priority detection targets?
- How can the overlap map be operationalised into detection rules and defensive controls?

3. Methodology

This project followed a four-phase threat hunting methodology grounded in structured intelligence collection, standardised framework mapping, and analytical correlation.

3.1 Phase 1 — Industry & Region Scoping

The engagement began with the definition of a precise threat landscape boundary. Industry selection was based on the critical infrastructure designation of the sector, its known attractiveness to nation-state actors (for espionage, disruption, or financial gain), and its geopolitical significance in the target region.

Region scoping considered geopolitical tensions, historical attack patterns against the sector, and known adversary presence as documented in open-source reporting. This scoping exercise ensured the subsequent APT search was targeted and operationally relevant rather than broadly theoretical.

3.2 Phase 2 — APT Identification via SOCRadar.io

SOCRadar.io was used as the primary threat intelligence aggregation and discovery platform. SOCRadar provides curated threat actor intelligence, mapping adversaries to industries, regions, motivation types, and known campaigns.

Using SOCRadar's threat actor search and filtering capabilities, a search was conducted against the defined industry and regional parameters. This yielded a corpus of 1,138 APT groups and threat actor clusters with documented or suspected activity matching the defined scope.

- Filter: Industry Vertical → [Selected Sector]
- Filter: Geographic Region → [APAC + Eastern Europe]
- Filter: Threat Actor Type → Nation-State & State-Sponsored
- Result: 1,138 threat actors returned

From this population, four threat actors were selected for deep-dive analysis based on their operational tempo, sophistication, documented attack history against the target sector, and availability of public TTP intelligence:

- APT41 (China-linked, dual espionage/financial motivation)
- KIMSUKY (North Korea-linked, espionage focus)
- SANDWORM (Russia-linked, destructive operations)
- LAZARUS GROUP (North Korea-linked, financial crime and espionage)

3.3 Phase 3 — TTP Analysis with MITRE ATT&CK

For each selected APT group, TTPs were researched using the MITRE ATT&CK knowledge base. MITRE ATT&CK provides a globally accessible, curated database of adversary behaviours mapped to a standardised taxonomy of 15 tactic categories and hundreds of techniques and sub-techniques.

Each APT's technique set was documented across all 15 tactic categories: Reconnaissance, Resource Development, Initial Access, Execution, Persistence, Privilege Escalation, Defense Evasion (Stealth), Defense Impairment, Credential Access, Discovery, Lateral Movement, Collection, Command and Control, Exfiltration, and Impact.

3.4 Phase 4 — ATT&CK Navigator Mapping & Overlap Analysis

The MITRE ATT&CK Navigator was used to create individual technique layers for each APT group, visualising their TTP coverage as a heat map across the ATT&CK matrix. Four individual layers were created (one per APT), and a composite overlap layer was then generated.

The overlap layer identifies techniques shared by multiple APT groups, colour-coded by the number of actors employing the same technique. Techniques appearing in 3 or 4 of the actor layers represent the highest-priority detection targets, as they reflect adversary-agnostic attack patterns that transcend individual group tooling or campaigns.

The resulting layer data has been exported and is included as Appendix A (layer_by_operation.xlsx) with full technique-level detail for all four APT groups and the composite overlap.

4. APT Group Profiles

The following profiles summarise the four APT groups selected for deep-dive analysis. Each profile covers nation-state attribution, primary targeting behaviour, motivations, and notable operations.

4.1 APT41 — Double Dragon

APT41 (Double Dragon / Winnti Group)

Nation-State Origin: People's Republic of China (PRC) — Ministry of State Security (MSS) linked

Primary Targets: Healthcare, Telecommunications, Technology, Gaming, Financial Services, Critical Infrastructure

Notable Operations: ShadowHammer supply chain attack (2019), multiple COVID-19 research institution breaches (2020), Citrix & Zoho exploitation campaigns

APT41 is uniquely distinguished by its dual mandate: conducting state-sponsored espionage on behalf of Chinese intelligence services while simultaneously running financially motivated cybercrime operations. This dual mission has resulted in one of the broadest known TTP profiles among tracked APT groups.

- Motivation: Espionage (IP theft, government intelligence) + Financial gain (ransomware, fraud)
- Primary tools: POISONPLUG, HIGHNOON, MESSAGETAP, custom RATs and rootkits
- Typical initial access: Spearphishing, supply chain compromise, exploitation of public-facing applications
- Known for: Operating simultaneously across espionage and financial objectives within the same campaigns

4.2 KIMSUKY

KIMSUKY (Velvet Chollima / Black Banshee)

Nation-State Origin: Democratic People's Republic of Korea (DPRK) — Reconnaissance General Bureau (RGB) linked

Primary Targets: South Korean Government, Think Tanks, Academics, UN Officials, Nuclear/Defence sectors globally

Notable Operations: Targeted South Korean nuclear energy policy researchers, operations against UN Panel of Experts members, BabyShark malware campaigns

KIMSUKY specialises in targeted intelligence collection operations, typically beginning with extensive reconnaissance and social engineering. The group is notable for its use of convincing spearphishing lures tailored to specific individuals, often impersonating academics, journalists, or policy researchers.

- Motivation: Strategic intelligence gathering in support of North Korean state interests
- Primary tools: BabyShark, AppleSeed, PEBBLEDASH, Gold Dragon

- Typical initial access: Highly targeted spearphishing emails, malicious documents, browser credential theft
- Known for: Long-dwell reconnaissance operations and targeted social engineering

4.3 SANDWORM

SANDWORM (Voodoo Bear / IRIDIUM)

Nation-State Origin: Russian Federation — GRU Unit 74455 (Main Centre for Special Technologies)

Primary Targets: Ukrainian Critical Infrastructure, Energy Sector, Government Networks, Olympic organisations, Media

Notable Operations: BlackEnergy attacks on Ukrainian power grid (2015, 2016), NotPetya wiper (2017), Olympic Destroyer (2018), Industroyer/CRASHOVERRIDE ICS malware

SANDWORM is considered one of the most destructive cyber threat actors ever documented. Its operations have caused real-world physical consequences including power outages affecting hundreds of thousands of civilians. The group's deployment of wiper malware and ICS-targeting capabilities makes it an extreme risk to critical infrastructure.

- Motivation: Geopolitical disruption, sabotage, psychological operations supporting Russian state objectives
- Primary tools: NotPetya, BlackEnergy, Industroyer, Olympic Destroyer, Cyclops Blink
- Typical initial access: Exploitation of network perimeter devices, spearphishing, supply chain attacks
- Known for: Destructive wiper campaigns, ICS/SCADA targeting, collateral damage at global scale

4.4 LAZARUS GROUP

LAZARUS GROUP (Hidden Cobra / Zinc / APT38)

Nation-State Origin: Democratic People's Republic of Korea (DPRK) — Lazarus Group / Bureau 121

Primary Targets: Global Financial Institutions, Cryptocurrency Exchanges, Defence Contractors, Energy Sector

Notable Operations: Sony Pictures hack (2014), Bangladesh Bank SWIFT heist (\$81M, 2016), WannaCry ransomware (2017), multiple cryptocurrency exchange compromises

LAZARUS GROUP is the primary financial cybercrime arm of the North Korean state, tasked with generating hard currency to circumvent international sanctions. The group combines sophisticated intrusion capabilities with aggressive financial theft operations, having stolen an estimated billions of dollars in cryptocurrency.

- Motivation: Financial theft (sanctions evasion), espionage, destructive operations
- Primary tools: BLINDINGCAN, HOPLIGHT, Manuscript, AppleJeus (macOS/Linux), custom SWIFT malware

- Typical initial access: Spearphishing, trojanised software, job offer social engineering lures
- Known for: Cryptocurrency heists, SWIFT network manipulation, cross-platform malware development

5. TTP Analysis by Tactic

The following section presents the complete TTP landscape documented across all four APT groups, organised by MITRE ATT&CK tactic category. This dataset was derived from MITRE ATT&CK Enterprise v19 and verified against public threat intelligence reporting for each actor.

RECONNAISSANCE

- Active Scanning
- Gather Victim Host Information
- Gather Victim Identity Information
- Gather Victim Network Information
- Gather Victim Org Information
- Phishing for Information
- Search Closed Sources
- Search Open Technical Databases
- Search Open Websites/Domains
- Search Threat Vendor Data
- Search Victim-Owned Websites

RESOURCE DEVELOPMENT

- Acquire Access
- Acquire Infrastructure
- Compromise Accounts
- Compromise Infrastructure
- Develop Capabilities
- Establish Accounts
- Generate Content
- Obtain Capabilities
- Stage Capabilities

INITIAL ACCESS

- Drive-by Compromise
- Exploit Public-Facing Application
- External Remote Services
- Phishing
- Replication Through Removable Media
- Supply Chain Compromise
- Trusted Relationship

- Valid Accounts
- Content Injection

EXECUTION

- Command and Scripting Interpreter
- Exploitation for Client Execution
- Hijack Execution Flow
- Native API
- Scheduled Task/Job
- User Execution
- Windows Management Instrumentation
- Poisoned Pipeline Execution
- Software Deployment Tools
- System Services
- Serverless Execution

PERSISTENCE

- Account Manipulation
- Boot or Logon Autostart Execution
- Boot or Logon Initialization Scripts
- Create Account
- Create or Modify System Process
- Event Triggered Execution
- External Remote Services
- Modify Authentication Process
- Modify Registry
- Scheduled Task/Job
- Server Software Component
- Valid Accounts
- Traffic Signaling

PRIVILEGE ESCALATION

- Access Token Manipulation
- Account Manipulation
- Boot or Logon Autostart Execution
- Create or Modify System Process
- Domain or Tenant Policy Modification
- Escape to Host
- Event Triggered Execution
- Exploitation for Privilege Escalation
- Process Injection
- Scheduled Task/Job
- Valid Accounts

DEFENSE EVASION (STEALTH)

- Access Token Manipulation
- Debugger Evasion
- Deobfuscate/Decode Files or Information
- Execution Guardrails
- Hide Artifacts
- Hijack Execution Flow
- Indicator Removal
- Masquerading
- Obfuscated Files or Information
- Pre-OS Boot
- Process Injection
- Reflective Code Loading
- Rootkit
- System Binary Proxy Execution
- Virtualization/Sandbox Evasion
- Valid Accounts

DEFENSE IMPAIRMENT

- Disable or Modify System Firewall
- Disable or Modify Tools
- Domain or Tenant Policy Modification
- Exploitation for Defense Impairment
- File and Directory Permissions Modification

- Modify Authentication Process
- Modify Registry
- Network Boundary Bridging
- Prevent Command History Logging
- Rogue Domain Controller
- Safe Mode Boot
- Subvert Trust Controls
- Weaken Encryption

CREDENTIAL ACCESS

- Adversary-in-the-Middle
- Brute Force
- Credentials from Password Stores
- Exploitation for Credential Access
- Forced Authentication
- Input Capture
- Modify Authentication Process
- Multi-Factor Authentication Interception
- Network Sniffing
- OS Credential Dumping
- Steal Application Access Token
- Steal or Forge Authentication Certificates
- Steal or Forge Kerberos Tickets
- Steal Web Session Cookie
- Unsecured Credentials

DISCOVERY

- Account Discovery
- Application Window Discovery
- Browser Information Discovery
- Domain Trust Discovery
- File and Directory Discovery
- Group Policy Discovery
- Network Service Discovery
- Network Share Discovery
- Network Sniffing
- Password Policy Discovery

- Permission Groups Discovery
- Process Discovery
- Remote System Discovery
- Software Discovery
- System Information Discovery
- System Network Configuration Discovery
- System Owner/User Discovery
- Virtualisation/Sandbox Evasion

LATERAL MOVEMENT

- Exploitation of Remote Services
- Internal Spearphishing
- Lateral Tool Transfer
- Remote Service Session Hijacking
- Remote Services
- Replication Through Removable Media
- Software Deployment Tools
- Taint Shared Content
- Use Alternate Authentication Material

COLLECTION

- Archive Collected Data
- Automated Collection
- Browser Session Hijacking
- Clipboard Data
- Data from Cloud Storage
- Data from Configuration Repository
- Data from Information Repositories
- Data from Local System
- Data from Network Shared Drive
- Data Staged
- Email Collection
- Input Capture
- Screen Capture
- Video Capture

COMMAND AND CONTROL (C2)

- Application Layer Protocol
- Data Encoding
- Data Obfuscation
- Dynamic Resolution
- Encrypted Channel
- Fallback Channels
- Hide Infrastructure
- Ingress Tool Transfer
- Multi-Stage Channels
- Non-Application Layer Protocol
- Non-Standard Port
- Protocol Tunneling
- Proxy
- Remote Access Tools
- Traffic Signaling
- Web Service

EXFILTRATION

- Automated Exfiltration
- Data Transfer Size Limits
- Exfiltration Over Alternative Protocol
- Exfiltration Over C2 Channel
- Exfiltration Over Other Network Medium
- Exfiltration Over Physical Medium
- Exfiltration Over Web Service
- Scheduled Transfer
- Transfer Data to Cloud Account

IMPACT

- Account Access Removal
- Data Destruction
- Data Encrypted for Impact
- Data Manipulation
- Defacement
- Disk Wipe
- Endpoint Denial of Service
- Financial Theft

- | |
|-----------------------------|
| • Firmware Corruption |
| • Inhibit System Recovery |
| • Network Denial of Service |
| • Resource Hijacking |
| • Service Stop |
| • System Shutdown/Reboot |

6. Cross-APT TTP Overlap Analysis

The MITRE ATT&CK Navigator overlap layer was created by stacking the individual technique layers for all four APT groups (APT41, KIMSUKY, SANDWORM, LAZARUS GROUP) and computing technique-level intersection. The resulting composite layer, included in the accompanying Excel file (layer_by_operation.xlsx), reveals the degree of TTP convergence across nation-state threat actors.

6.1 Tactic-Level Overlap Matrix

The table below shows the tactic-level TTP coverage for each APT group and the resulting overlap score. A 4/4 overlap score indicates that all four groups employ techniques within that tactic category:

TTP Category	APT41	KIMSUKY	SANDWORM	LAZARUS	Overlap Score
Reconnaissance	✓	✓	✓	✓	4/4
Resource Development	✓	✓	✓	✓	4/4
Initial Access	✓	✓	✓	✓	4/4
Execution	✓	✓	✓	✓	4/4
Persistence	✓	✓	✓	✓	4/4
Privilege Escalation	✓	✓	✓	✓	4/4
Defense Evasion (Stealth)	✓	✓	✓	✓	4/4
Defense Impairment	✓	✓	✓	✓	4/4
Credential Access	✓	✓	✓	✓	4/4
Discovery	✓	✓	✓	✓	4/4
Lateral Movement	✓	✓	✓	✓	4/4
Collection	✓	✓	✓	✓	4/4
Command & Control	✓	✓	✓	✓	4/4
Exfiltration	✓	✓	✓	✓	4/4
Impact	✓	✓	✓	✓	4/4

The 4/4 overlap score across all 15 tactic categories is a significant finding: it indicates that APT41, KIMSUKY, SANDWORM, and LAZARUS GROUP all operate across the full ATT&CK kill chain. This is consistent with their classification as highly mature, full-lifecycle threat actors capable of conducting complete intrusion operations from initial access through to impact.

6.2 High-Priority Overlapping Techniques

Within the tactic categories, specific techniques emerge as particularly high-overlap — appearing across all four APT groups. These represent the highest-priority detection targets for any organisation facing nation-state-level threats:

Credential Access — Highest Overlap Techniques

- OS Credential Dumping — All four groups employ credential dumping as a standard post-exploitation step
- Steal Web Session Cookie — Session token theft observed across all four actors
- Brute Force — Credential spraying and brute force used by all groups during initial reconnaissance phases
- Multi-Factor Authentication Interception — Increasingly prevalent across all four actors

Defense Evasion — Highest Overlap Techniques

- Masquerading — Renaming tools to mimic legitimate system binaries, seen universally
- Obfuscated Files or Information — Custom encoding and packing of malicious payloads
- Indicator Removal — Log clearing and artefact wiping standard across all four groups
- Virtualization/Sandbox Evasion — Anti-analysis techniques to defeat automated sandboxes

Command & Control — Highest Overlap Techniques

- Encrypted Channel — All four groups use TLS/SSL encrypted C2 channels
- Application Layer Protocol — HTTP/HTTPS-based C2 used by all actors
- Proxy — Multi-hop proxy infrastructure to obscure true C2 origins
- Web Service — Legitimate cloud services (GitHub, OneDrive, Dropbox) abused for C2

Persistence — Highest Overlap Techniques

- Boot or Logon Autostart Execution — Registry run keys and startup folders used universally
- Scheduled Task/Job — Windows Scheduled Tasks for persistent execution across all groups
- Valid Accounts — Compromise and abuse of legitimate credentials for long-term access

6.3 Navigator Layer Interpretation

The composite overlap layer in the MITRE ATT&CK Navigator (exported in `layer_by_operation.xlsx`) uses colour intensity to represent overlap frequency. The scoring logic applied was:

- Score 1 (light): Technique used by one APT group
- Score 2 (medium): Technique shared by two APT groups
- Score 3 (dark): Technique shared by three APT groups

- Score 4 (darkest): Technique shared by all four APT groups — highest priority detection target

The individual sheets in the workbook represent the raw ATT&CK Enterprise v19 technique matrix as it applies to each actor (APT41, KIMSUKY, SANDWORM, LAZARUS), plus a composite 'layer by operation' sheet representing the overlap.

7. Key Findings & Risk Assessment

7.1 Critical Findings

F-01	Full Kill-Chain Coverage: All four APT groups demonstrate capability and intent to operate across all 15 ATT&CK tactic categories, indicating mature, full-lifecycle threat actors capable of conducting end-to-end intrusion campaigns without external dependency.
F-02	Universal TTP Convergence: The 4/4 overlap across all tactic categories reveals that nation-state actors from China, Russia, and North Korea — despite distinct geopolitical objectives and tooling — converge on the same fundamental attack patterns. This convergence enables adversary-agnostic detection strategies.
F-03	Credential Theft as Universal Entry Point: All four groups prioritise credential theft (OS Credential Dumping, web session cookie theft, MFA interception) as a primary post-exploitation objective, indicating that identity-based defences are the single most impactful defensive investment.
F-04	Living-off-the-Land (LotL) Dominance: The heavy overlap in Defense Evasion techniques — particularly Masquerading, System Binary Proxy Execution, and Trusted Developer Utilities Proxy Execution — reflects a sector-wide shift to LotL techniques that defeat signature-based detection.
F-05	Encrypted C2 Universal: All four groups use encrypted, covert C2 channels leveraging legitimate protocols and services (HTTPS, WebSockets, cloud storage APIs), making network-layer detection reliant on behavioural analysis rather than content inspection.

7.2 Risk Rating

Based on the TTP overlap analysis, organisations in the target industry and region face nation-state-level threat actor activity across the full ATT&CK matrix. The consolidated risk rating is:

CRITICAL Overall Threat Rating	Threat Actor Sophistication: Nation-State (Tier 1) Attack Surface Exposure: High (full kill-chain capability) Detection Complexity: High (LotL, encrypted C2)
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8. Defensive Recommendations

The following recommendations are derived directly from the TTP overlap analysis and are prioritised by the frequency with which techniques appear across the four APT groups. Defences addressing 4/4 overlap techniques deliver the broadest risk reduction.

8.1 Priority 1 — Identity & Credential Hardening

- Deploy Privileged Access Workstations (PAWs) for all administrative activity to reduce credential exposure.
- Implement phishing-resistant MFA (FIDO2/WebAuthn) for all external and privileged internal access.
- Deploy credential monitoring solutions (e.g., Have I Been Pwned integration, Dark Web monitoring via SOCRadar) to detect compromised credentials before adversarial use.
- Enable Windows Credential Guard to protect LSA secrets from OS Credential Dumping techniques.
- Implement Just-in-Time (JIT) and Just-Enough-Access (JEA) for privileged accounts.

8.2 Priority 2 — Detection Engineering (LotL Focus)

- Deploy Endpoint Detection & Response (EDR) with behavioural detection to identify process injection, LSASS access, and masquerading activity independent of signature matches.
- Enable comprehensive PowerShell logging (ScriptBlock logging, Module logging, Transcription) to capture obfuscated script execution.
- Implement SIEM correlation rules targeting the highest-overlap techniques: scheduled task creation by non-admin accounts, Base64-encoded PowerShell execution, LSASS memory access.
- Deploy Windows Defender Application Control (WDAC) or AppLocker to restrict Living-off-the-Land Binary (LOLBin) abuse.

8.3 Priority 3 — Network & C2 Detection

- Deploy encrypted traffic inspection (TLS inspection) at network egress points to enable C2 channel detection without relying on content signatures.
- Implement DNS monitoring and threat intelligence integration to detect Dynamic Resolution and domain generation algorithm (DGA) patterns.
- Use network behaviour analytics to baseline legitimate cloud service usage and detect anomalous exfiltration over legitimate web services.
- Deploy honeypots and deception technology to detect lateral movement and internal reconnaissance activity.

8.4 Priority 4 — Threat-Informed Detection Development

- Map existing detection rules to the ATT&CK overlap layer to identify coverage gaps in the organisation's detection posture.
- Develop and test detection rules specifically targeting the 4/4 overlap techniques identified in Section 6.2.
- Conduct regular purple team exercises simulating the TTP profiles of all four APT groups, using this report as the adversary emulation baseline.
- Subscribe to SOCRadar.io threat actor feeds for the target industry/region to maintain current awareness of new TTPs and campaigns.

9. Tools & References

9.1 Tools Used

SOCRadar.io

Open-source threat intelligence and attack surface management platform used for APT discovery and filtering. Provided the corpus of 1,138 APTs relevant to the defined industry and region.

<https://socradar.io>

MITRE ATT&CK

The MITRE ATT&CK knowledge base — a globally accessible, standardised framework of adversary tactics, techniques, and procedures (TTPs) based on real-world observations.

<https://attack.mitre.org>

MITRE ATT&CK Navigator

A web-based tool for visualising, annotating, and comparing ATT&CK technique coverage. Used to create individual APT layers and the composite overlap layer.

<https://mitre-attack.github.io/attack-navigator>

9.2 Key References

- MITRE ATT&CK Enterprise v19 — <https://attack.mitre.org/versions/v19/>
- APT41 Profile — MITRE ATT&CK Group G0096
- KIMSUKY Profile — MITRE ATT&CK Group G0094
- SANDWORM Profile — MITRE ATT&CK Group G0034
- LAZARUS GROUP Profile — MITRE ATT&CK Group G0032
- SOCRadar Threat Actor Intelligence — <https://socradar.io/resources/threat-actors>
- Mandiant APT41 Report: A Dual Espionage and Cyber Crime Operation (2019)
- US-CERT Alert AA20-301A: North Korean Advanced Persistent Threat Focus: Kimsuky
- CISA Advisory AA22-110A: Russian State-Sponsored and Criminal Cyber Threats to Critical Infrastructure
- US-CERT Alert AA21-048A: Lazarus Group Malware Analysis

9.3 Appendices

- Appendix A: layer_by_operation.xlsx — MITRE ATT&CK Navigator export containing individual technique layers for APT41, KIMSUKY, SANDWORM, and LAZARUS GROUP, plus the composite overlap layer.
- Appendix B: SOCRadar.io search parameters and filter configuration used for APT corpus generation.

End of Report

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